

⑥ Inference (Generating New Content)

The prompt you give AI, it looks at your input and calculates the most statistically likely next word, repeats until process is complete.

It refines by using models (or feedback loops and human guidance to ensure the generated answers remain helpful, safe and accurate.

Examples

→ GitHub Copilot

It acts as an AI pair programmer that suggests and auto-completes lines or whole blocks of computer code inside your editor. It analyzes the code you are writing and uses generative AI models to predict what code you need next.

→ Midjourney

It generates detailed digital images, art and realistic photos from text descriptions. It uses diffusion models (AI that start with random noise and refines it step by step) to turn words into visual graphics.

→ ChatGPT

It creates text, answers questions, writes essays, and helps with brainstorming. It uses LLM to predict and write natural human like responses based on your typed prompts.

17. Training dataset is a collection of data used to teach a machine learning or artificial intelligence model how to recognise patterns, make predictions or perform specific tasks.

A larger, cleaner and more diverse training dataset helps the model perform accurately on new real world data.